G.C.E. (O/L) · MATHEMATICS ENGLISH MEDIUM

QUESTION PAPER

Model Paper — 05

Mathematics I

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Time: 2 hours **Total marks:** 100

Index number: .....

Important:

- This question paper has two parts: Part A and Part B. Answer **all** questions on this paper itself.
- Use only the space provided under each question for your working and your answer.
- Show the relevant steps and write the correct units.
- Part A: 2 marks for each question. Part B: 10 marks for each question.
- Scrap paper may be obtained for calculations.

For examiner's use only						
Part A (1–25)	**B 1**	**B 2**	**B 3**	**B 4**	**B 5**	**Total**
/ 50	/ 10	/ 10	/ 10	/ 10	/ 10	/ 100

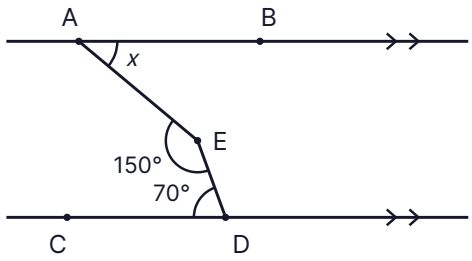
Part A

Answer all questions on this paper itself. (2 marks each; take π = 22/7 where needed)

1. Six workers can level a plot in Nimesha's garden in 20 days. All the workers work at the same rate. How many more workers must be employed so that the work is finished in 8 days?

Answer: .....

2. In the figure, AB ∥ CD, ∠AED = 150° and ∠EDC = 70°. Find the size of ∠BAE (x).



Answer: .....

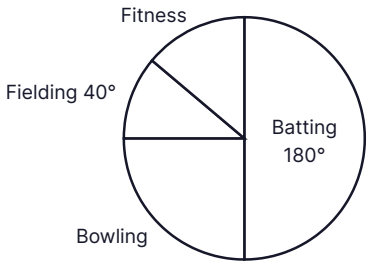
3. Simplify: $\frac{1}{2x} + \frac{3}{5x} - \frac{3}{10x}$

Answer: .....

4. The base of a right prism is a triangle. Which one of the following can **not** be a face of this prism? Put a tick (✓) against the correct answer.
(i) a rectangle (ii) a square (iii) a trapezium

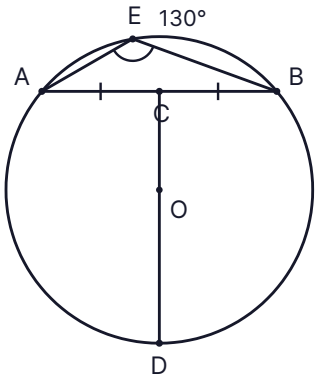
Answer: .....

5. The pie chart shows the favourite training activity of the members of a cricket club. 36 members chose batting. Find the number of members who chose fielding.



Answer: .....

6. In the circle with centre O, DOC is a straight line and C is the mid-point of the chord AB. ∠AEB = 130°. Find the size of ∠CDB.



Answer: .....

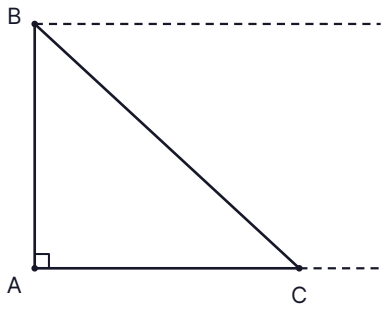
7. Factorise: $3x^2 - 48$

Answer: .....

8. It is given that $10^{0.6021} = 4$. Find the value of lg 400.

Answer: .....

9. In the figure, AB is a vertical post on horizontal ground. A bird is sitting at B and a cat is at C on the ground; ∠BAC = 90°. The dashed lines are horizontal. On the figure, mark the angle of elevation of the bird from the cat as a and the angle of depression of the cat from the bird as b, and write the relationship between a and b.



Answer: .....

10. A cylindrical flower pot has height equal to 5 times its base radius r. Find, in its simplest form, the ratio of the area of paper needed to wrap its curved surface to the area of paper needed to cover its circular base. (Curved surface area of a cylinder = $2\pi rh$)

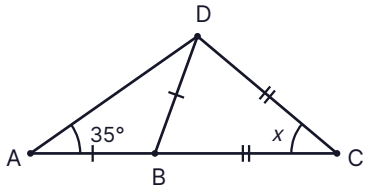
Answer: .....

11. Kavindu rolls two fair tetrahedral dice together. Each die has four faces numbered 2, 3, 5, 7. Find the probability that the sum of the numbers on the two bottom faces is 10.

Answer: .....

12. In the figure, ABC is a straight line and ∠DAB = 35°. Sides marked with the same number of ticks are equal in length. Find the size of x.

Answer: .....



13. Let ξ = {x : $1 \leq x \leq 10$, x is an integer}, A = {prime numbers in ξ} and B = {multiples of 3 in ξ}. Put a tick (✓) against the **correct** statement.
(i) $A \cap B = \{3\}$ (ii) $A \cup B = \{2, 3, 5, 6, 7\}$ (iii) $n(A') = 5$

Answer: .....

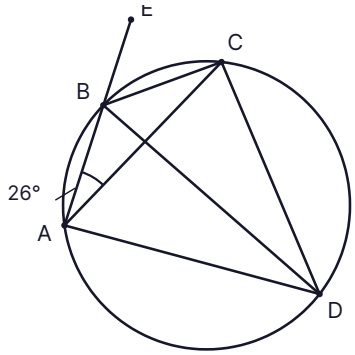
14. Simplify: $(9a^4b \times 4a^2b^3) \div 6a^3b^2$

Answer: .....

15. In a geometric progression the second term is −40 and the third term is −20. Find the fifth term.

Answer: .....

16. In the figure, ABCD is a cyclic quadrilateral and BD bisects ∠ADC. AB is produced to E. If ∠BAC = 26°, find the size of ∠CBE.



Answer: .....

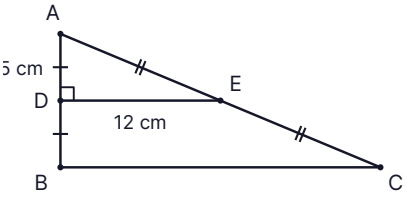
17. Find the highest common factor (HCF) of $12a^2b^3$, $18a^3b^2$ and $30ab^4$.

Answer: .....

18. The base of a right triangular prism of height h has an area of 2464 cm^2. A cylinder of the same height h has the same volume as the prism. Find the base radius r of the cylinder. (Volume of a cylinder = $\pi r^2 h$)

Answer: .....

19. In triangle ABC, D and E are the mid-points of AB and AC respectively. ∠ADE = 90°, AD = 5 cm and DE = 12 cm. Find the perimeter of the trapezium BCED.

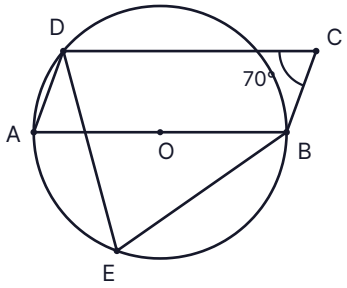


Answer: .....

20. Find the equation of the straight line passing through the points (2, 7) and (5, 1).

Answer: .....

21. In the figure, O is the centre of the circle and AOB is a straight line. A, B, D and E are points on the circle. ABCD is a parallelogram and ∠BCD = 70°. Find the size of ∠DEB.

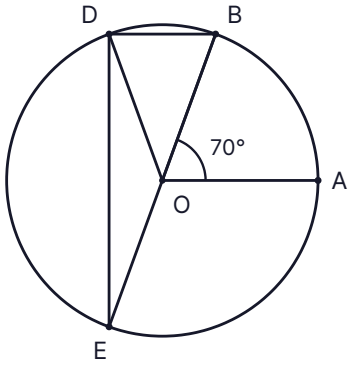


Answer: .....

22. Sachini invested a sum of money for 2 years at 4% per annum compound interest, with interest added at the end of each year. The interest for the second year was Rs. 2 600 and the total amount at the end of the 2 years was Rs. 67 600. Find the sum she invested.

Answer: .....

23. In the figure, O is the centre of the circle and OA ∥ BD. ∠AOB = 70°. Find the size of ∠BED.

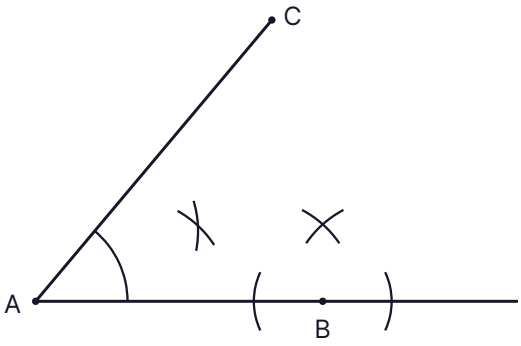


Answer: .....

24. Solve the inequality $4x + 15 > 3$ and write the negative integer values of x that satisfy it.

Answer: .....

25. In the figure, the line AB touches a circle at B and the line AC also touches the same circle. Part of the construction needed to find the centre O of the circle is shown. Using your knowledge of constructions, complete the construction and mark the centre O.



Answer: .....

Part B

Answer all questions on this paper itself. (10 marks each)

1. Ruwan is a three-wheeler driver. He pays income tax at 20% on the part of his monthly income that exceeds Rs. 75 000. After paying the tax, he spends $\frac{1}{5}$ of the money he receives on fuel, and then $\frac{3}{8}$ of the remainder on other expenses.

(i) What fraction of the money he receives is left after spending on fuel? (Marks 1)

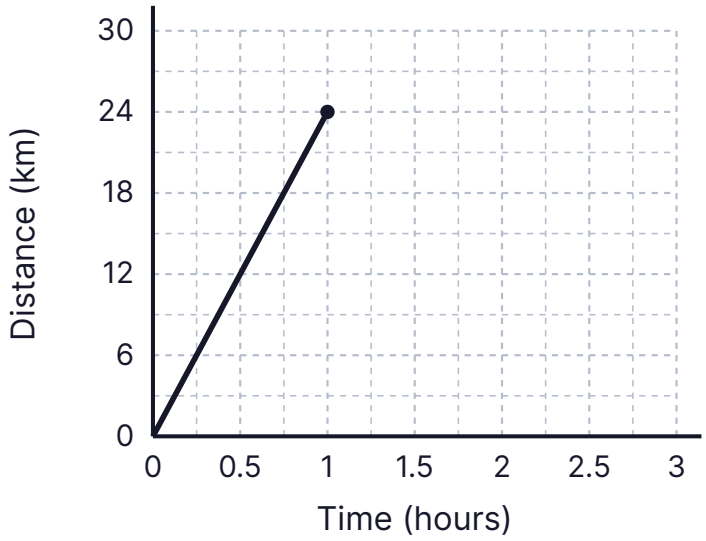
(ii) What fraction of the money he receives is left after spending on fuel and on other expenses? (Marks 2)

(iii) The money left after all these is Rs. 47 500. Find separately the amount spent on fuel and the amount spent on other expenses. (Marks 2)

(iv) Find Ruwan's monthly income before tax. (Marks 3)

(v) Suppose Ruwan had been exempt from income tax, and the amount he spends on fuel remained the same as in part (iii). Express the amount spent on fuel as a percentage of his monthly income. (Marks 2)

2. **(a)** Kavindu cycles from his home to the cricket ground at a uniform speed, taking 1 hour to cover the 24 km. He practises at the ground for 1 hour, and then cycles back home along the same route at a uniform speed, taking 45 minutes. The first part of the distance–time graph of his journey is drawn on the grid.



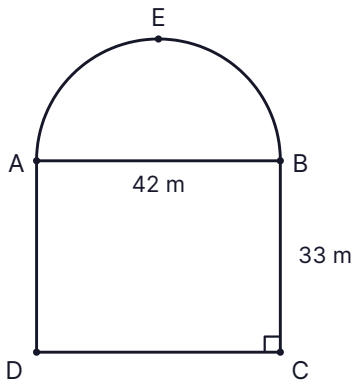
(i) Complete the distance–time graph of his journey. (Marks 2)

(ii) Find his speed on the return journey in km/h. (Marks 2)

(b) The figure shows a garden of Nimesha. It consists of a rectangular part ABCD and a semicircular flower bed AEB with AB as its diameter. AB = 42 m and BC = 33 m. Use $\pi = 22/7$.

(i) Find the perimeter of the whole figure (the boundary of the garden). (Marks 3)

(ii) The area of the rectangle ABCD is how many times the area of the semicircle AEB? (Marks 3)



3. Sachini rents out a house at Rs. 15 000 per month and collects the rent for the whole year in one payment. The assessed annual value of the house is Rs. 120 000 and the municipal council charges annual rates of 15% of the assessed value. She spends 10% of the year's rent on maintenance of the house.

(i) Find the annual rates she has to pay. (Marks 2)

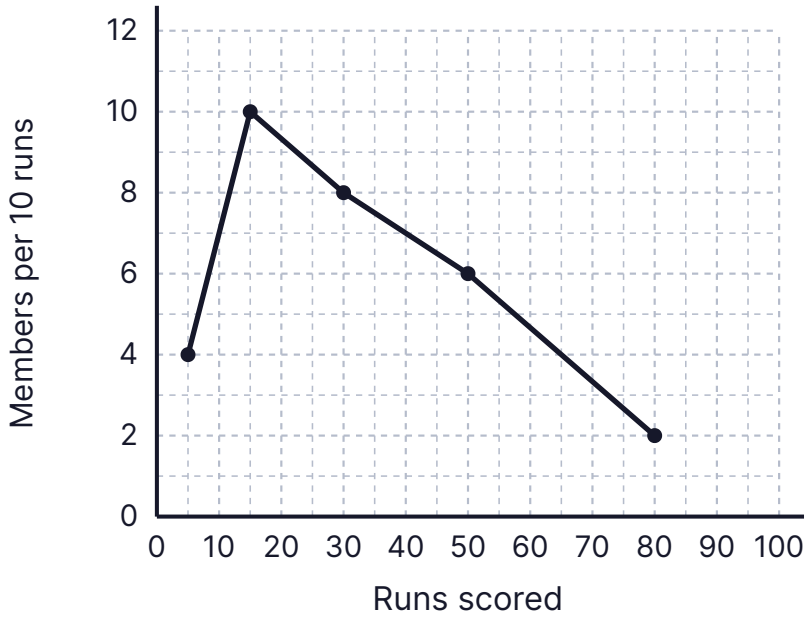
(ii) Find the money she has left after paying the rates and the maintenance cost. (Marks 2)

(iii) She adds some more money to this amount and buys shares of Rs. 80 each. Each share pays an annual dividend of Rs. 6, and she receives Rs. 12 000 as dividends for the year.

(a) Find the number of shares she bought and the total amount she invested in shares. (Marks 3)

(b) Find the extra money she added, as a percentage of the total amount invested. (Marks 3)

4. The frequency polygon shows the runs scored by 50 members of a cricket club in a tournament. Its points are plotted at the mid-values of the classes, and the vertical axis shows the number of members **per 10 runs** of class width. (A class such as 20 – 40 includes 20 but not 40.) An incomplete table is also given.



Runs scored	Number of members	Cumulative frequency
0 – 10	4	4
10 – 20		
20 – 40		
40 – 60		
60 – 100		

(i) Using the polygon, complete the "Number of members" column of the table. (Take care: the classes do not all have the same width.) (Marks 3)

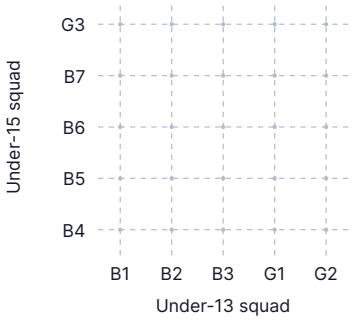
(ii) On the same grid, draw the histogram for the above information. (Marks 3)

(iii) Complete the cumulative frequency column of the table. (Marks 2)

(iv) A member is chosen at random. Find the probability that this member scored 40 runs or more. (Marks 2)

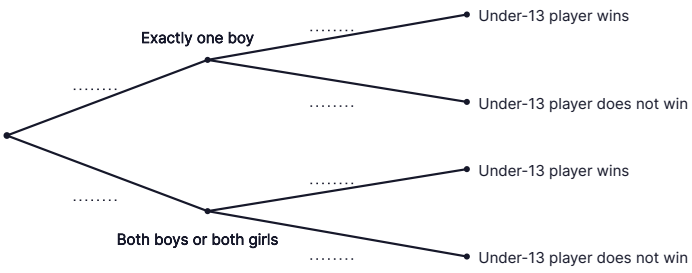
5. Kavindu coaches a cricket club. Its Under-13 squad has five players: three boys B1, B2, B3 and two girls G1, G2. Its Under-15 squad has five players: four boys B4, B5, B6, B7 and one girl G3. One player is selected at random from each squad to take part in a challenge. A grid to show the outcomes is given.

(i) Mark all the possible outcomes of the selection on the grid with the symbol 'X'. (Marks 2)



(ii) Using closed curves, enclose the outcomes in which exactly one of the two selected players is a boy. Hence find the probability of this event. (Marks 3)

(iii) In the challenge, the probability that the player from the Under-13 squad wins is $\frac{1}{5}$ if exactly one of the two selected players is a boy, and $\frac{1}{2}$ if both are boys or both are girls. Complete the probabilities on the tree diagram. (Marks 3)



(iv) Find the probability that the player from the Under-13 squad does **not** win. (Marks 2)

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